

SATELLITE TRANSMISSION

INTERMEDIATE DATA RATE (IDR) EQUIPMENT IN C BAND

GENERIC REQUIREMENTS (No. GR/IDR-01/04.NOV2006)

(Supersedes GR No. GR/IDR-01/03. JUN 2001)

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**DEPARTMENT OF TELECOMMUNICATIONS
GOVERNMENT OF INDIA
TELECOMMUNICATION ENGINEERING CENTRE
KHURSHID LAL BHAVAN, JANPATH
NEW DELHI 110001
INDIA**

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HISTORY SHEET

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Intermediate Data Rate (IDR) Equipment operating in C band.	G/IDR-01/01.MAR 92	First Release of GR
Intermediate Data Rate (IDR) Equipment operating in C band.	G/IDR-01/02.FEB 95	Second Release of GR
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REFERENCES

All the standards are subject to revision. The versions/issues of the standards indicated in this GR, or their subsequent versions/editions shall be applicable.

TEC GRs/IRs

TEC standard no. SD/EMI-02/02.SEP 2001 with Amendment No.1 1 dated 30.5.2003 and Corrigendum dated 6.5.2005.

QA DOCUMENTS

- QM 115 (1997) - Reliability Methods and Predictions
- QM 118 - Quality and Reliability in product Design
- QM 205 - Guidelines for Standard of Workmanship for Printed Boards
- QM 206 - Guidelines for Standard of Workmanship for Printed Board Assemblies
- QM 210 - Guidelines for Standard of Workmanship for Surface Mounted Devices
- QM 301 - Transmission Equipment General Documentation.
- QM 333 (1990) – Specification for environmental testing of electronic equipments for transmission and switching use, for operation, transportation and storage.

ITU RECOMMENDATIONS

- ITU-R S.522-5, S.614-4, S.579-6 ITU-T V.36, G.703, G.712

OTHER STANDARDS

- ISO 9001:2000
- IS 8437 {1993}- Guide on the effects of current passing through the human body
- IS 13252 {2003}- Safety of information technology equipment including electrical business equipment
- IS 12063 (1987) - Classification of degree of protection provided by enclosures of the electrical equipment
- IS 10437 (1986) – Safety requirements of radio transmitting equipments.
- CISPR 22 {2003}- Limits and methods of measurement of radio disturbance characteristics of Information Technology Equipment.
- IEC 61000-4-2 {2001}- Testing and measurement techniques of Electrostatic discharge immunity test.
- IEC 61000-4-4 -{1995 with Amendment 1 (2000) and Amendment 2 (2001)} Testing and measurement techniques of electrical fast transients/burst immunity test.
- IEC 61000-4-3 -{2002} Radiated RF Electromagnetic Field Immunity test.
- IEC 61000-4-5 - {2001} Test & Measurement techniques for Surge immunity tests.
- IEC 61000-4-6 -{2001} Immunity to conducted disturbances, induced by radio frequency fields.
- IESS 308 Rev.11, 310 Rev.3

PART 1

TECHNICAL REQUIREMENTS

GR No. GR/IDR-01/04 NOV2006

1.0 SCOPE

This document provides the generic characteristics of Intermediate Data Rate (IDR) equipment for operating with all types of earth stations working in INSAT system using modulation switchable between QPSK and TCM-8PSK with options for information rate as $N \times 64$ kbps (where $N=1$ to 32), 2.048 Mbps, 8.448 Mbps and 34.368 Mbps. The design of IDR carriers shall be based upon the ITU-R Recommendations S.522-5, S.614-4 and S.579-6. These carriers may share transponders with carriers employing other approved modulation techniques and/or with other IDR carriers. The occupied satellite bandwidth unit for these carriers is equal to $0.7(\text{QPSK})/0.467(\text{TCM-8 PSK})$ times the transmission rate inclusive of guard band between adjacent carriers. The specifications of Up/Down converters forming part of the IDR Equipment have also been provided. The system shall be configured with redundancy of (1:1) for remote station and (N:1) for main station where N can be upto 7.

1.1 FUNCTIONAL AND TECHNICAL REQUIREMENTS

1.1.1 FUNCTIONAL REQUIREMENTS

- 1.1.1.1 IDR equipment shall be designed to comply with ITU-R Recommendations S.522-5, S.614-4 and S.579-6.
- 1.1.1.2 The IDR Equipment shall support the concatenation of Reed-Solomon (RS) code. The shortened systematic Reed-Solomon Codes are listed in Table-1.
- 1.1.1.3 Since errors from the decoder tend to occur in bursts, block interleaving shall be employed in conjunction with the RS outercode to reduce the correlation among erroneous symbols in a RS Code word. The interleaving details shall be as per IESS 308 Rev.11 and IESS 310 Rev.3 documents.
- 1.1.1.4 For IDR Carrier with rate 2 Mb/s, the RS Encoder/ Interleaver shall be inserted after the 96 kb/s ESC overhead framing unit and before the FEC encoder. On the receive side (demodulator) the RS decoder/de-interleaver shall be inserted after the FEC decoder and before the 96 kb/s ESC overhead deframing unit.
- 1.1.1.5 It shall be possible to enable or disable the use of RS outer coding from the front-panel control of the IDR channel unit.
- 1.1.1.6 Since the transmission rate and FEC used for IDR modulator and demodulator may be different for the transmit (modulator) and receive (demodulator) paths, it shall be possible to independently select the same or different RS codecs for RS encoder and decoder respectively in order to be compatible with the transmission rates (**hence RS code words) being used by the IDR modulator and demodulator, irrespective of the inner FEC rate (1/2 or 3/4 with QPSK modulation and 2/3 with TCM-8PSK modulation) being used.**
- 1.1.1.7 Programmability of the RS Codec shall be provided to allow maximum flexibility and ease of use in selecting the different RS Codes corresponding to the various IDR information rates.

1.1.1.8 Synchronization and scrambling requirements shall be as per IESS 308 Rev.11 and IESS 310 Rev 3.

1.1.1.9 When using the RS outer coding, the following shall be provided:

- a) An indication of the composite rate (i.e., information rate plus overhead) E_b/N_0 at the input to the RS decoder. The E_b/N_0 reading shall be derived from the inner FEC modem and shall have an accuracy of equal to or better than $\pm 0.5\text{dB}$ over a range extending from the point at which the inner FEC modem loses synchronization to a maximum value of at least 10 dB. Design objectives are: an E_b/N_0 measurement accuracy of $\pm 0.25\text{ dB}$ (or better) for E_b/N_0 values less than or equal to 10 dB and a maximum E_b/N_0 readout of 15dB ($\pm 1\text{ dB}$ accuracy, for $10\text{dB} < E_b/N_0 \leq 15\text{dB}$).
- b) A "low E_b/N_0 " alarm to signify the onset of degraded RS decoder performance. The activation of this alarm shall only trigger appropriate audible and visual indicators. It shall be possible to adjust the trigger level for this alarm over an E_b/N_0 range of 5 dB to 7 dB in incremental steps of 0.25 dB.
- c) The BER at the input to the RS decoder shall be displayed over a range of about 10^{-2} to at least 10^{-9} .
- d) The BER at the output to the RS decoder shall be displayed over a range of 10^{-4} to at least 10^{-10} . This BER value shall be derived from the input BER versus output BER performance characteristic of the RS decoder (e.g., by using a look-up table of the transfer characteristic).

1.2 TECHNICAL REQUIREMENTS

1.2.1 Up Converter

- a) Output Frequency : 5.925 to 6.425 GHz
- b) Frequency sense : No inversion
- c) Input Frequency : $70 \pm 18\text{ MHz}$
- d) Input level range : -35 dBm to -15 dBm
- e) Input Impedance : 75 Ohms Unbalanced
- f) Input Return loss : 23 dB minimum
- g) Input Connector type : BNC Female
- h) Frequency Selection : By Synthesizer with 125 KHz step size.
- i) Output Impedance : 50 Ohms Unbalanced
- j) Output Return loss : 20 dB minimum

k) 1 dB compression point	: 0 dBm
l) Output Connector type	: SMA (Female)
m) Output Level Range	: -20 dBm to 0 dBm
n) Output Level stability	: ± 0.25 dB/day Over $25^{\circ}\text{C} \pm 10^{\circ}\text{C}$ with constant Input
o) IF-RF Gain	: 15 dB minimum
p) Gain Response	: ± 0.50 dB/36 MHz
q) Gain adjustment	: 20 dB in steps of 0.5dB
r) Inter modulation products	: -30 dBc for two carriers at a level of -3 dBm. each at output.
s) AM/PM Conversion	: $0.2^{\circ}/\text{dB}$
t) Group delay	: Linear : 0.05 ns/MHz. Parabolic : $0.005 \text{ ns}/\text{MHz}^2$. Ripple : 1 ns p-p/ ± 18 MHz
u) Spurious output:	
i) Signal related	: -50dBc
ii) Signal independent	: -70dBm at minimum attenuation
iii) Harmonics	: -25dBc
v) Frequency Stability	: ± 0.1 ppm/day at a constant temperature of 25°C
w) Phase Noise	: -55 dBc/Hz at 100 Hz offset -70 dBc/Hz at 1 KHz offset -80 dBc/Hz at 10 KHz offset -90 dBc/Hz at 100 KHz offset -100 dBc/Hz at 1 MHz offset The single side band phase noise due to both the continuous and spurious components integrated over the bandwidth of 10 Hz to $0.3R$ Hz away from the centre frequency. where R is the transmission rate in bits/s, which should not exceed 2 deg rms.

1.2.2 Down Converter

a) Input Frequency	: 3.7 GHz to 4.2 GHz
b) Frequency sense	: No inversion
c) Output Frequency	: 70 ± 18 MHz

- d) Input Level Range : -60 dBm to -20 dBm
- e) Input Impedance : 50 Ohms Unbalanced
- f) Input Return loss : 20 dB Minimum
- g) Input Connector Type : SMA Female
- h) Frequency Selection : By frequency synthesizer with 125 kHz step size.
- i) Output Impedance : 75 Ohms Unbalanced
- j) Output Return loss : 23 dB Minimum
- k) 1 dB compression point : + 5 dBm
- l) Output Connector Type : BNC Female
- m) Output Level Range : -35 dBm to +5 dBm
- n) Output Level Stability : ± 0.25 dB/Day over $25^{\circ}\text{C} \pm 10^{\circ}\text{C}$
- o) RF /IF Gain : 45 dB Minimum
- p) Gain Response : ± 0.50 dB/36 MHz
- q) Gain adjustment range : 20 dB in steps of 0.5dB
- r) Inter modulation Products : -40dBc for two carriers at a level of -3 dBm each at output.
- s) AM/PM Conversion : $0.2^{\circ}/\text{dB}$
- t) Group delay : Linear : 0.05 ns/MHz
Parabolic : $0.005 \text{ ns}/\text{MHz}^2$
Ripple : 1ns peak to peak / ± 18 MHz
- u) Spurious :
 - i) Signal related : -50dBc
 - ii) Signal independent : -70dBm at minimum attenuation
 - iii) Harmonics : -25dBc
- v) Frequency Stability : ± 0.1 ppm/day at a constant temperature of 25°C
- w) Phase Noise : -55 dBc/Hz at 100 Hz offset
-70 dBc/Hz at 1 KHz offset
-80 dBc/Hz at 10 KHz offset
-90 dBc/Hz at 100 KHz offset
-100 dBc/Hz at 1 MHz offset
The single side band phase noise due to both the

continuous and spurious components integrated over the bandwidth of 10 Hz to $0.3R$ Hz away from the center frequency where R is the transmission rate in bits/s, should not exceed 2 deg rms.

- x) Noise Figure : 15 dB Maximum
- y) Image Rejection : 70 dB
- z) Monitoring & Alarms : Power ON Indicator
REF frequency monitoring, Los phase lock.
Summary Fail alarm, Potential free contact

1.2.3 Transmit System and Satellite Equalizer (Not mandatory for 2 Mbps IDR)

- a) Input Centre Frequency : 70 MHz
- b) Input Impedance : 75 Ohms/BNC
- c) Input Return Loss : 20 dB minimum
- d) Input Level : 0 to -20 dBm
- e) Output Centre Frequency : 70 MHz
- f) Output Level : 0 to -20 dBm
- g) Transmit Equalization over 70 ± 18 MHz:
 - i) Amplitude : Slope ± 2.5 dB.
Parabolic upto 1 dB/MHz^2
 - ii) Group Delay : -6ns to + 11ns
- h) Satellite Group Delay : Linear Slope: 0 to $\pm 1 \text{ ns/MHz}$
over 70 ± 18 MHz : Parabolic : 0 to 25 ns/MHz^2

1.2.4 Modulator:

- a) IF input Range : $70 \text{ MHz} \pm 18 \text{ MHz}$, step size 22.5 kHz
- b) Information Rate : i) $N \times 64 \text{ kbps}$ (where $N=1$ to 32)
Note: Option to be indicated by the purchaser. ii) 2.048 Mbps
iii) 8.448 Mbps
iv) 34.368 Mbps
- c) Scrambler : As per ITU-T Recommendation V.36
- d) Coding : Convolutional
- e) Rate : Switchable between $1/2$ (QPSK), $3/4$ (QPSK) and $2/3$ (TCM-8PSK).

Inner Code	: Convolution Encoding / Pragmatic TCM encoding
Outer Code	: Reed-Solomon Code (see Table 1).

- | | |
|---------------------------|---|
| f) Modulation | : Switchable between QPSK and TCM-8PSK |
| g) IF Interface | : BNC female, 75 ohms unbalanced |
| h) Output return loss | : > 18 dB |
| i) Output level | : 0 dBm |
| j) Level Adjustability | : 0 to 20 dB varied through the front panel key board in steps of 0.5 dB. |
| k) Level Stability | : ± 0.2 dB/day at constant temperature of 25°C. |
| l) On/Off Levels | : 50 dB minimum |
| m) Output Spurious Levels | : -40 dBc/4 kHz for rates less than and equal to 2.048 Mbps
-50 dBc/4 kHz for rates above 2.048 Mbps |
| n) Phase accuracy | : $\pm 2^\circ$ |
| o) Amplitude accuracy | : ± 0.2 dB |

1.2.5 Demodulator:

- | | |
|--|--|
| a) IF Input Range | : 70MHz $\pm$ 18 MHz, step size 22.5 kHz |
| b) Output Data | : i) N*64 kbps (where N=1 to 32)
ii) 2.048 Mbps
iii) 8.448 Mbps
iv) 34.368 Mbps |
| Note: Option to be indicated by the purchaser. | |
| c) Descrambler | : As per ITU-T Recommendation V.36 |
| d) Decoding | : Viterbi / TCM (inner)
Reed-Solomon (outer) (see Table 1) |
| e) Rate | : Switchable between 1/2 (QPSK), 3/4 (QPSK) and 2/3 (TCM-8PSK). |
| f) Demodulation | : Switchable between QPSK and TCM-8PSK |
| g) IF Interface | : BNC female, 75 ohms unbalanced |
| h) I/P return loss | : > 18 dB |
| i) Input level | : -45 dBm to -35 dBm |

j) Bit Error Ratio (IF loop back):

The Eb/No is referred to the modulated carrier power and to the composite data rate (information rate + overhead) entering the RS Encoder. Eb/No values include a 0.3 dB allowance for increased inter-symbol interference (ISI) caused by allowing the use of analog modem filters whose designs are based on the transmission rate without the use of RS outer coding.

i) With RS Coding:

	Composite Data rate Eb/No (dB)		
BER better than	Rate 1/2 (QPSK)	Rate 3/4 (QPSK)	Rate 2/3 (TCM)
10^{-4}			6.1 dB
10^{-6}	4.1 dB	5.6 dB	6.5 dB
10^{-7}	4.2 dB	5.8 dB	6.7 dB
10^{-8}	4.4 dB	6.0 dB	6.9 dB
10^{-10}	5.0 dB	6.3 dB	7.3 dB

For transmission over the satellite an RF link degradation of 0.4 dB for 10^{-4} , 10^{-6} , 10^{-7} and 10^{-8} relative to the modems IF loop back performance is needed to account for possible degradation to phase noise, imperfect amplitude/group delay equalisation, adjacent channel interference (ACI) and transponder non-linearities. At a BER of 10^{-10} , the IF-RF degradation for rate 3/4 FEC is 0.7 dB, for rate 1/2 FEC, 0.9 dB and for 2/3, 0.6 dB.

ii) Without RS Coding:

Due to the steepness of the BER versus Eb/No response curve when using the Reed-Solomon outer coding, an inordinately long period of time is necessary to detect a sufficient number of errors to determine the BER performance with reasonable degree of confidence at even moderate Eb/No values. Assuming that the Reed-Solomon outer codec is functioning, determining the BER performance of the codec without the Reed-Solomon outer coding, i.e., the performance of the inner code only, would enable users to quickly determine whether or not the modem is functioning correctly. As a guideline, the following BER versus Eb/No performance (IF loopback) is suggested.

	Composite Data rate Eb/No (dB)		
BER better than	Rate 1/2 (QPSK)	Rate 3/4 (QPSK)	Rate 2/3 (TCM)
10^{-3}	4.2 dB	5.3 dB	6.3 dB
10^{-4}			7.3 dB
10^{-5}			8.2 dB
10^{-6}	6.1 dB	7.6 dB	9.0 dB
10^{-7}	6.7 dB	8.3 dB	9.8 dB
10^{-8}	7.2 dB	8.8 dB	10.4 dB
10^{-10}	9.0 dB	10.3 dB	

1.2.6 Buffers

a) Minimum Buffer Size : 1.5 ms to 19.5 ms as per table below

Satellite orbit inclination (degrees)	Buffer capacity (ms) for various circuit configurations			
	Case 1	Case 2	Case 3	Case 4
0.1(Nominal)	1.5	2.4	2.7	No buffer required
0.5	2.5	4.4	4.7	
1.0	3.9	7.4	7.7	
1.5	5.5	10.4	10.7	
2.0	6.9	13.2	13.5	
2.5	8.5	16.4	16.7	
3.0	9.9	19.2	19.5	

Notes:

Case 1 : Digital network at both ends.

Case 2 : One end of the link remotely timed from other end.

Case 3 : Recovered timing from one station is used for transmitting to another network

Case 4: No synchronous digital network at either end and no buffer requirements

b) Terrestrial Frame length : 125 μ s to 2 ms

c) Slip Size : Multiples of terrestrial frame length.

d) Control : From front panel

e) Indicators : To be viewed from front panel & remote.

1.2.7 Timing Accuracy : 2.048 Mbps/2 MHz digital signal in both directions of transmissions shall be derived in one of the following ways:

a) From a clock with an accuracy of 1×10^{-11}

b) From an incoming clock received from a remote earth station by satellite. In this case the remote earth station must derive timing by method (a) above

c) In cases where there is no synchronous digital network at either end, but the channels are converted to analog voice circuits, the internal clock of the PCM MUX equipment is of an accuracy of about 50×10^{-6} .

- d) As an emergency backup a local clock with a long term stability of **atleast 1×10^{-5} per month shall be used.**

1.2.8 Terrestrial Interfaces:

N*64 kbps (N=1 to 32)	: ITU-T Recommendation G.703 120 ohms symmetric pair
2.048 Mbps	: ITU-T Recommendation G.703 120 ohms symmetric pair
8.448 Mbps	: ITU-T Recommendation G.703 75 ohms BNC, unbalanced
34.368 Mbps	: ITU-T Recommendation G.703 75 ohms BNC, unbalanced
SNMP V.2	: for NMS

1.2.9 Redundancy

Switchover Modes	: Independent for receive/ transmit chains Automatic with lockout override Manual Remote or Local
Monitoring	: Standby modem continually sampled for status
Display	: Front panel display of relevant status
Remote Mode	: Via serial link
Power Supplies	: Redundant
Multitransponder operation	: Optional for upto four transponders with transponder hopping switch.

1.3 Monitoring & Controls

1.3.1 Common to all modules: Lamp test (Lights all indicators)

Reset (Causes full restart of microprocessor).

1.3.2 Modulator	: TX Channel Frequency TX Power Level continuous in steps of 0.5 dB Differential Encoder, ON/OFF Scrambler, ON/OFF Baseband loopback, ON/OFF
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TX Carrier, ON/OFF
 Force switch over in 1:1 redundancy mode,
 Normal/Force
 Transmit CW (Continuous Wave: no phase
 transitions),
 ON/OFF
 1/0 Sequence Generator, Normal/Sequence
 ($2^{11}-1$) Sequence Generator, Normal/PN Sequence

1.3.3 Demodulator

: RX channel frequency

IF loopback, ON/OFF

Baseband loopback, ON/OFF

1.3.4 Decoder/Buffer Module

: Plesiochronous Buffer Bypass, ON/OFF

Plesiochronous Buffer Frame Length

Plesiochronous Buffer Re-center

Differential decoder, ON/OFF

FEC decoder rate, 1/2 or 3/4 for QPSK-IDR
 and 2/3 for TCM-IDR

Descrambler, ON/OFF

(Above functions to be set from modem front panel).

1.3.5 Engineering Service Channel

: Voice
 Single/Multidestination (optional)
 Duplex Mode

1.4 Indicators

1.4.1 Common to all modules

Power ON/OFF : Signals all voltages present

Online : Signifies active unit for redundant operation

Local : Manual control

Loopback : Test mode indication/warning

1.4.2 Demodulator

Dem Lock : Demod loops are locked and tracking

Carrier Lock : Carrier loop is locked

BTL Lock : Bit timing loop is locked

Synth Lock : Receive synthesizer locked

1.4.3 Modulator

Carrier On : Modulator healthy and transmitting

Clk Gen Lock : Clock Generator loops locked

TX Syn Lock : TX Synthesizers locked

1.4.4 Decoder/Buffer Module

DBM Syn : Decoder in sync

Decoder Syn : Viterbi decoder synchronised

Buffer Center : Buffer re-center occurred

1.4.5 Upconverter : Power on indicator

1.4.6 Down converter : Power on indicator

1.5 Alarms :

1.5.1 Common to all modules:

Fault : Signals a module fault

1.5.2 Demodulator:

IFL Fault : Loss of IF input

1.5.3 Modulator:

Input Loss : Input data from interface lost

1.5.4 Decoder/Buffer Module:

Frame Slip : Buffer over/under flow occurred

T1 Alarm : Interface module generated alarm

1.5.5 Upconverter : Summary fail alarm

1.5.6 Down converter : Summary fail alarm

1.6 Engineering Service Channel (ESC) :

- a) Type : Digital
- b) Overhead Rate : 96 kbps for data rate at 2.048 Mbps and above.

- c) Capacity : Two channels of 32 kbps for voice & voice band data; One channel of 8 kbps for data (RS-422 Interface) and Four alarms
- d) Features : Pass band of 0.3 to 3.4 kHz with calling facility on either buzzer or ringer with visual indication (with reset facility) and handset. Potential free contact for extension of buzzer/ringer calling facility to a centralised place shall be provided. Contact points for extension of audio alarm shall also be provided.
- e) Nominal 4W Trans/ receive line up level at monitoring points (600 Ohms) : – 3 dBm preferable. Level adjustment facility to be provided. The range of adjustment may be indicated. Suitable access for line up shall also be provided.
- f) V.F. response : As per ITU-T Recommendation G-712.
- g) Clock Rate : 1 kHz (Derived from incoming data stream normally).
- h) Backup clock : 1 kHz with 1×10^{-5} per month. (in case of failure of normal clock.)
- i) Frame length : 125 μ s

1.7 MTBF: 50,000 hours minimum

1.8 MARKING: Equipment shall be marked suitably as under

- a. Name of manufacturer
- b Year of manufacturing
- c Frequency of operation

PART 2

GENERAL REQUIREMENTS
GR No. GR/IDR-01/04. NOV2006

2.0 GENERAL REQUIREMENTS

2.1 Engineering requirements

- 2.1.1 The joints in the equipment shall have protection as per BIS standard IS 12063 {1987}” Classification of degree of protection provided by enclosures of the electrical equipment” {equivalent to IEC-529} to meet at least the protection level of IP-65.
- 2.1.2 The equipment should follow proper construction practice to minimise unintended radiations due to leakage from any gap or monitoring points. All unused if/rf ports and monitoring points should be terminated. The power flux density shall not exceed 1 mW/cm^2 at a distance of 2.5 cms.
- 2.1.3 The equipment shall adopt state-of-the-art technology.
- 2.1.4 The equipment shall be of compact and composite construction and lightweight. The actual dimensions and weight shall be furnished by the manufacturer.
- 2.1.5 All connectors shall be reliable and of standard type to ensure for over 500 failure-free mating operations under the environmental conditions specified.
- 2.1.6 All connectors and the cable used shall be of low loss type and suitably shielded.
- 2.1.7 The equipment shall be housed in standard 19” (600 mm) rack or slim rack with a maximum height of 2250 mm and with front access. For outdoor equipment (or part of equipment), construction practice shall be followed as specified.
- 2.1.8 The equipment shall preferably work on natural cooling. However, the use of in-built DC operated fans is permitted in some equipment. An alarm shall be extended to indicate failure of fan/fans. MTBF of each fan shall be more than 50,000 hours.
- 2.1.9 The supervisory indications and other control switches shall be provided at convenient locations on the bay preferably at a height around 1500 mm for ease of maintenance.
- 2.1.10 The plug-in units shall be hot swappable to allow their removal/insertion while the equipment is in energized condition.
- 2.1.11 The mechanical design and construction of each/unit shall be inherently robust and rigid under all conditions of operation, adjustment, replacement, storage and transport and conforming to para 12 of BSNL QA document No. QM-333 {latest issue} “Specification for environmental testing of electronic equipments for transmission and switching use”.
- 2.1.12 Each sub-assembly shall be clearly marked with schematic reference to show its function, so that it is identifiable from the layout diagram in the handbook.
- 2.1.13 Each terminal block and individual tags shall be numbered suitably with clear identifying code and shall correspond to the associated wiring drawings.
- 2.1.14 All controls, switches, indicators etc. shall be clearly marked to show their circuit designation and functions.

- 2.1.15 Important Do's and Don'ts about the operation of the equipment shall be clearly indicated at a convenient place on the equipment.

2.2 OPERATIONAL REQUIREMENTS

- 2.2.1 The manufacturer shall guarantee the satisfactory performance of the equipment without any degradation in performance up to at an altitude upto 3,600 meters.
- 2.2.2 The equipment should be able to work without any degradation in performance in saline atmosphere near coastal areas and should be protected against corrosion.
- 2.2.3 Visual indication to show power ON/OFF status shall be provided.
- 2.2.4 Suitable visual indications shall be provided. (It is suggested that green colour for healthy and red colour for unhealthy conditions may be provided. Some other colour may be used for non urgent alarms).
- 2.2.5 The software/hardware in equipment shall not pose any problem due to changes in date and time caused by events such as changeover leap year etc. in the normal functioning of the equipment.
- 2.2.6 The equipment shall be provided with the proper dummy load connecting facilities, cooling arrangements, input/output, isolation /monitoring ports, as required.

2.3 QUALITY REQUIREMENTS

- 2.3.1 The manufacturer shall furnish the MTBF value. MTBF shall meet the values specified in GR. The calculations shall be based on the guidelines in either BSNL QA document no. QM-115 {1997} "Reliability Methods and Predictions" or any other international standard.
- 2.3.2 The equipment shall be manufactured in accordance with International quality management system ISO 9001:2000 for which the manufacturer should be duly accredited. A quality plan describing the quality assurance system followed by the manufacturer would be required to be submitted.

And

The equipment shall be manufactured as per the latest BSNL QA Guidelines indicated in Quality Manuals QM 118 {Quality and Reliability in product Design}, QM 205 {Guidelines for Standard of Workmanship for Printed Boards}, QM 206 {Guidelines for Standard of Workmanship for Printed Board Assemblies}, QM 210 {Guidelines for Standard of Workmanship for Surface Mounted Devices} and QM 301 {Transmission Equipment General Documentation}.The supplier shall furnish a certification from the manufacturer to this effect, which shall be verified at the time of technical specifications evaluation.

2.4 ENVIRONMENTAL REQUIREMENTS FOR EQUIPMENT

- 2.4.1 The equipment shall conform to the requirements for the Category B2 of the Environment Categories specified in BSNL QA document QM-333 {1990} "Specification for environmental testing of electronic equipments for transmission

and switching use" for operation, transportation and storage, including vibration and corrosion (salt mist).

2.5 MAINTENANCE REQUIREMENTS

- 2.5.1 Maintenance philosophy is to replace faulty units/subsystems after quick on-line analysis through monitoring sockets, alarm indications and Built-in Test Equipment. The actual repair will be undertaken at centralised repair centers. The corrective measures at site shall involve replacement of faulty units/sub-systems.
- 2.5.2 The equipment shall have easy access for servicing and maintenance.
- 2.5.3 Suitable alarms shall be provided for identification of faults in the system and faulty units.
- 2.5.4 Suitable provision shall be made for extension of summary alarms.
- 2.5.5 Ratings and types of fuses used are to be indicated

2.6 POWER SUPPLY

- 2.6.1 The equipments shall be fed from the station power plant. The equipment shall meet the following requirements in respect of the power supply:-
- 2.6.2 Nominal power supply is -48Volts DC, with a variation over the range -44Volts to -57V. The equipment should operate over this range without any degradation in performance.
- 2.6.3 The equipment shall be protected in case of voltage variation beyond the above specified range and also against input reverse polarity. The manufacturer shall furnish the value of guaranteed input voltage upto which no irreversible damage to the equipment will occur.
- 2.6.4 The actual power consumption shall be furnished by the manufacturer.
- 2.6.5 The derived DC voltages in the equipment shall have protection against over voltages, short circuit and overload.
- 2.6.6 The equipment shall have the option of operating from two independent sources of input power supply.
- 2.6.7 The equipment requiring AC mains shall operate from AC power supply without any degradation in the performance
Nominal AC voltage for Single Phase shall be 230 V with variation of -15% to +10%, at 50 ± 2 Hz.
Nominal AC voltage for Three Phase shall be 415 V with variation of -15% to +10%, at 50 ± 2 Hz.

2.7 ACCESSORIES

- 2.7.1 The supplier shall provide one complete set of:

- a) All the necessary interfaces, connectors, connecting cables and accessories required for satisfactory and convenient operation of the equipment. Types of connectors, adapters to be used and the accessories of the approved quality shall be clearly indicated in the operating manuals which should be in conformity with the detailed list in the GR.
 - a) Software and the arrangements to load the software at site.
- 2.7.2 Special tools, extender boards, extender cables and accessories essential for installation, operation and maintenance of the equipment shall be clearly indicated and supplied along with the equipment.
- 2.7.3 Special tools, extender boards, extender cables and accessories essential for repair of the equipment shall be clearly indicated and supplied in case the same are ordered.

2.8 DOCUMENTATION

2.8.1 TECHNICAL LITERATURE

- 2.8.1.1 Technical literature in English or Hindi with complete layout, detailed block schematic and circuit diagram of various assemblies with test voltages/waveforms at different test points of the units shall be provided. All aspects of installation, operation, maintenance and repair shall be covered in the manuals. The soft copy as well as hard copy of the manuals shall also be provided. The manuals shall include the following manuals.

2.8.2 INSTALLATION, OPERATION AND MAINTENANCE MANUAL

- a. Safety measures to be observed in handling the equipment;
- b. Precautions for installation, operation and maintenance;
- c. Test jigs and fixtures required, and procedures for routine maintenance, preventive maintenance, troubleshooting and subassembly replacement;
- d. Illustration of internal and external mechanical parts.
- e. Detailed description about the operation of the software used in the equipment including its installation, loading, debugging, etc.

2.8.3 REPAIR MANUAL

- a. List of replaceable parts used, including their sources and the approving authority;
- b. Detailed ordering information for all the replaceable parts shall be listed in the manual to facilitate reordering of spares as and when required;
- c. Procedure with flowchart for troubleshooting and subassembly replacement shall be provided. Test fixtures and accessories required for repair shall also be indicated. Systematic troubleshooting charts (fault-tree) shall be given for the probable faults with their remedial actions.

2.9 PROTECTION REQUIREMENTS

- 2.9.1 The equipment shall have a terminal for grounding the rack.
- 2.9.2 Protection against short circuit/open circuit in the accessible points shall be provided.

- 2.9.3 All switches/controls on front panel shall have suitable safeguards against accidental operation.
- 2.9.4 The equipment shall be adequately safeguarded to prevent entry of dust, insects and lizards.

2.10 SAFETY REQUIREMENTS

- 2.10.1 The operating personnel should be protected against shock hazards as per IS 8437 {1993} "Guide on the effects of current passing through the human body" [equivalent to IEC publication 60479-1{1984}].
- 2.10.2 The equipment shall conform to IS 13252 {2003} "Safety of information technology equipment including electrical business equipment" [equivalent to IEC publication 60950 {2001}] and IS 10437 {1986} "Safety requirements of radio transmitting equipments" [equivalent to IEC publication 60215].

2.11 ELECTROMAGNETIC COMPATIBILITY (EMC)

- 2.11.1 The equipment shall conform to the EMC requirements as per the following standards and limits indicated therein. A test certificate and test report shall be furnished from test agency which should be an accredited one, and details of accreditation shall be submitted. Following test results should be furnished:-
- 2.11.2 Conducted and radiated emissions:- To comply with Class A of CISPR 22 {2003} "Limits and methods of measurement of radio disturbance characteristics of Information Technology Equipments";
- 2.11.3 Electrostatic discharge:- To comply with IEC 61000-4-2 {2001} "Testing and measurement techniques of Electrostatic discharge immunity test" under following test levels:
- a. Contact discharge level 2 { ± 4 kV};
 - b. Air discharge level 3 { ± 8 kV};
- 2.11.4 Fast transients common mode (burst):- To comply with IEC 61000-4-4 {1995 with Amendment No.1 (2000) and Amendment No.2 (2001)} "Testing and measurement techniques of electrical fast transients/burst immunity test" under Level 2 {1 kV for DC power lines; 1 kV for signal control lines};
- 2.11.5 Immunity:- IEC 61000-4-3 {2002} "Radiated RF Electromagnetic Field Immunity test" under Test level 2 {test field strength of 3 V/m}for general purposes in frequency range 80 MHz to 1000 MHz and under test level 3 (10 V/m) for protection against digital radio telephones in frequency ranges 800 MHz to 960 MHz and 1.4 GHz to 2.0 GHz.
- 2.11.6 Surges line to earth coupling and line to line coupling:- To comply with IEC 61000-4-5 {2001}"Test & Measurement techniques for Surge immunity tests" under test levels of 0.5 kV for line to line coupling and 1 kV for line to earth coupling;
- 2.11.7 Radio frequency common mode:- To comply with IEC 61000-4-6 {2001} "Immunity to conducted disturbances, induced by radio frequency fields" under the test level 2 {3

V r.m.s.}; Clamp injection method for DC lines and Signal Control lines.

- 2.11.8 For tests for checking compliance to above EMC requirements, the method of measurements shall be in accordance with IEC/CISPR and TEC standard no. SD/EMI-02/02.SEP.2001 (with all amendments) and the references mentioned therein. Alternatively, corresponding relevant Euro Norms of the above IEC/CISPR standards are also acceptable subject to the condition that above mentioned frequency ranges and test levels are met as per TEC standard no. SD/EMI-02/02 SEP.2001 (with all amendments). The details of IEC/CISPR and corresponding Euro Norms are as under

IEC/CISPR	Euro Norm
CISPR 22	EN 55022
IEC 61000-4-2	EN 61000-4-2
IEC 61000-4-3	EN 61000-4-3
IEC 61000-4-4	EN 61000-4-4
IEC 61000-4-5	EN 61000-4-5
IEC 61000-4-6	EN 61000-4-6

TABLE-1**i) REED-SOLOMON CODES FOR QPSK-IDR**

Information Rate (kb/s)	RS Code (n, k, t)¹	Bandwidth Expansion [(n/k)-1]	Inter leaving depth	Maximum² RS Codec delay (ms)
64	(126, 112, 7)	0.125	4	115
256	(126, 112, 7)	0.125	4	029
384	(126, 112, 7)	0.125	4	019
704	(126, 112, 7)	0.125	4	010
2048	(219, 201, 9)	0.0896	4	007
8448	(194, 178, 8)	0.0899	4	0<2
34368	(208, 192, 8)	0.0833	4	0<2

ii) REED-SOLOMON CODE FOR TCM-IDR

Type of Service	Recommended information Rates (kbit/s)	RS Code (n,k)¹	Bandwidth Expansion [(n/k)-1]	Inter-leaving Depth (I_d)	Maximum² RS Codec Delay (ms)
Small TCM/IDR	512 1024	(219,201)	0.0896	4	19 10
TCM/IDR (With 96 kbit/s overhead)	2048 8448 34368	(219,201)	0.0896	8	8 2 <1

- Notes: 1. n = code length, k = information symbols, and t = symbol error correcting capability.
2. Design objective

TABLE - 2

For E3 IDR, the bandwidth occupied for Rate $\frac{1}{2}$ QPSK (with and without RS coding) exceeds 36MHz, hence, for 34.368 Mbps, only Rate $\frac{3}{4}$ QPSK and Rate $\frac{2}{3}$ 8PSK options should be mentioned. Info rate: $34.368 + 0.096 = 34.464$ Mbps.

Scheme	Transmission rate (Mbps)	Symbol rate (Mbps)	Occupied bandwidth MHz (alpha = 0.35)
$\frac{1}{2}$ QPSK	68.928	34.464	46.53
$\frac{3}{4}$ QPSK	45.952	22.976	31.02
$\frac{2}{3}$ TCM-8PSK	51.696	17.232	23.26
RS (208, 192) $\frac{1}{2}$ QPSK	74.672	37.336	50.40
RS (208, 192) $\frac{3}{4}$ QPSK	49.781	24.89	33.60
RS (219/201) $\frac{2}{3}$ TCM-8PSK	56.325	18.775	25.35

TABLE – 3**Bandwidth Calculations for IDR carriers**

Data Rate in Mbps	Scheme	Transmission Rate in Mbps	Symbol Rate in Mbps	Occupied Bandwidth MHz (alpha = 0.35)
2.048 + 96 Kbps ESC data	1/2QPSK	4.288	2.144	2.8944
	RS (208,192) 1/2 QPSK	4.645	2.3225	3.135
	3/4QPSK	2.859	1.4295	1.9299
	RS (208,192) 3/4 QPSK	3.096	1.548	2.089
	2/3 TCM-8PSK	3.126	1.072	1.447
	RS (219/201) 2/3 TCM- 8PSK	3.504	1.168	1.577
8.448 + 96 Kbps ESC data	1/2QPSK	17.088	8.544	11.534
	RS (208,192) 1/2 QPSK	18.512	9.256	12.496
	3/4QPSK	11.392	5.696	7.690
	RS (208,192) 3/4 QPSK	12.341	6.17	8.33
	2/3 TCM-8PSK	12.816	4.272	5.767
	RS (219/201) 2/3 TCM- 8PSK	13.963	4.654	6.283
34.368+96 Kbps ESC data	1/2QPSK	68.928	34.914	47.134
	RS (208,192) 1/2 QPSK	74.672	37.336	50.404
	3/4QPSK	45.952	22.976	31.018
	RS (208,192) 3/4 QPSK	49.781	24.890	33.602
	2/3 TCM-8PSK	51.696	17.232	23.26
	RS (219/201) 2/3 TCM- 8PSK	56.325	18.775	25.34
STM1 CARRIER				
155.52MBPS	8 PSK O.9 TPC	172.8	57.6	70.272(1.22 TIMES SR)

NOTE :

1. Transmission Rate = Data Rate * Inverse of FEC Rate
2. T.R with RS Coding = Value at 3 * 208/192 for QPSK ,
Value at 3 * 219/201 for TCM 8PSK
3. Symbol Rate= Transmission Rate/2 for QPSK.
Transmission Rate/3 for TCM 8PSK
4. Spreading factor of 1.2 times the symbol rate is used for filters.
5. A typical value of 0.15 times symbol rate is used as guard band..

6. Occupied Bandwidth is 1.35 times Symbol Rate
7. The bandwidth of the carrier at the -3.8 dB points is approx the same as the symbol rate.
8. The bandwidth of the carrier at the -12 dB points is approx 1.28 times the symbol rate. The occupied bandwidth, i.e. spacing between carriers is approx 1.35 times the symbol rate
9. The allocated bandwidth of the carrier at the -40 dB points is approx 1.4 times of the symbol Rate

No. of E1 carriers per 36 MHz Transponder in different modulation schemes

	QPSK RATE $\frac{1}{2}$ RS ON	QPSK RATE $\frac{1}{2}$ RS OFF	QPSK RATE $\frac{3}{4}$ RS ON	QPSK RATE $\frac{3}{4}$ RS OFF	8 PSK RATE $\frac{2}{3}$ RS ON	8 PSK RATE $\frac{2}{3}$ RS OFF
Eb/N0 in dB @ 1E-6 BER	4.1	6.1	5.6	7.35	6	8.6
Power limited	40	24	26	16	24	12
Bandwidth limited	10	12	16	18	22	24
Actual No. of carriers	10	12	16	16	22	12

Overhead of 96 kbps added to the basic rate for all data rates. Link calculations done for C band transponder EIRP 34 dBW, G/T -5 dB/K

**No. of carriers per 36MHz transponder in different modulation schemes
(Bandwidth limited case)**

	QPSK RATE $\frac{1}{2}$ RS ON	QPSK RATE $\frac{1}{2}$ RS OFF	QPSK RATE $\frac{3}{4}$ RS ON	QPSK RATE $\frac{3}{4}$ RS OFF	8 PSK RATE $\frac{2}{3}$ RS ON	8 PSK RATE $\frac{2}{3}$ RS OFF
E1	10	12	16	18	22	24
E2	2	3	4	4	5	6
E3	0	0	1	1	1	1

Annex for the reference of the purchaser

(To form part of the GR)

- 6.1 The clauses contained in this Annex are for the reference of the purchaser only and these clauses shall not be tested during Type Approval testing.
- a. As and when software bugs are found/determined, the manufacturer shall provide patches/firmware replacement if involved as mutually agreed between purchaser of equipment and supplier. Modified documentation wherever applicable shall also be supplied.
 - b. The manufacturer/supplier shall furnish the list of recommended spares.
 - c. The supplier shall have maintenance/repair facility in India.
 - d. The purchaser shall have the option to call for the equipment which shall work upto an altitude of 3600 meters without any degradation in the performance”.

ABBREVIATIONS

ADPCM	Adaptive Differential Pulse Code Modulation
AM	Amplitude modulation
BER	Bit Error Ratio
BNC	Bayonet nut coupling
BSNL	Bharat Sanchar Nigam Limited
cm	centimeter
°C	degree Celsius
CISPR	International Special Committee on Radio Interference
DC	Direct current
dB	decibel
dBc	decibels referenced to the carrier level
DCME	Digital circuit multiplication equipment
dBm	decibels referenced to a milli Watt
Eb/No	Energy per bit/Noise spectral density

EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
ESC	Engineering Service Channel
FEC	Forward Error Correction
GHz	Giga Hertz
GR	Generic Requirements
Hz	Hertz
IEC	International Electrotechnical Commission
IESS	Intelsat Earth Station Standards
IF	Intermediate Frequency
IFL	Intermediate Frequency loss
INSAT	Indian Satellite System
IP	International protection : designator for degrees of protection in conjunction with two numerals
IS	Indian Standard
ISO	International Standardisation Organisation
ITU	International Telecommunication Union

kbps	kilo bits per second
kHz	kilo Hertz
kV	kilovolt
LO	Local oscillator
Mbps	Mega bits per second
MHz	Mega Hertz
mm	milli meter
ms	milli second
mW	milli watt
MTBF	Mean Time Between Failure
MUX	Multiplexing
ns	nano second
p-p	peak to peak
PCM	Pulse Code Modulation
PM	Phase modulation
PSK	Phase shift keying
QA	Quality Assurance

QM	Quality Manual
QPSK	Quadrature phase shift keying
REF	Reference
RF	Radio Frequency
rms	Root mean square
RS	Reed-Solomon
RX	Receive
SMA	Subminiature Adapter
TCM	Trellis-Coded Modulation
TX	Transmit
V	Volts
V/m	volts/meter
W	Watt
μ s	micro second